COMBINATORIAL EULER-SIEVE IDENTITIES, HADAMARD–POISSON FLUX DETECTION, AND CONDITIONAL NODAL-WALL IDENTITIES FOR RIEMANN'S Ξ-FUNCTION

THIJS MISSIAEN

Abstract. We analyze three constructions for Riemann's completed zeta function $\Xi(t-\mathrm{i}d) := \xi(\frac{1}{2}+d+\mathrm{i}t)$ and its canonical transverse Wronskian $W(d,t) := \frac{1}{2}\partial_d|\Xi(t-\mathrm{i}d)|^2 = 16\int_0^\infty y\sinh(2dy)I(y,t)\,\mathrm{d}y$, separating exact identities from structural obstructions. First, we prove the finite cubic-window Dirichlet identity for $\zeta(s)\prod_{p\le X}(1-p^{-s})$ on $1 \le N < (\lfloor X\rfloor+1)^3$, while showing that $F_2(z) = 1+z+\frac{1}{2}z^2$ vanishes at $z=-1\pm\mathrm{i}$, $\mathcal{P}_{>X}(s)$ diverges in $0<\sigma\le 1$, and Kronecker density on small primes produces arbitrary complex excursions. Second, we evaluate the Poisson–Hadamard lobe integral $\mathcal{F}_-(d,d_*) = -\arcsin(d/d_*)$ and Sylvester inertia jump on a pole-free entire two-zero perturbation as contrapositives of Lagarias's equivalence. Third, we formulate the Cauchy–Riemann nodal-wall identity along $I(d,\tau(d)) = 0$ with explicit non-circularity hypotheses and contrast Jacobi's $\vartheta_3(0\mid \mathrm{i}e^{4u}) \neq 0$ with the imaginary root $\Phi(\mathrm{i}\eta_0) = 0$ at $\eta_0 = 0.319415\cdots < \frac{\pi}{8}$.

1. Introduction

Let $\Xi(t-\mathrm{i}d) := \xi(\frac{1}{2}+d+\mathrm{i}t) = R(d,t)+\mathrm{i}I(d,t)$ denote Riemann's completed zeta function at $s=\frac{1}{2}+d+\mathrm{i}t$ [12, 14], represented by the even Jacobi-theta integral

$$\Xi(t-\mathrm{i}d) = 8\int_0^\infty \Phi(u)\cos\big(2(t-\mathrm{i}d)u\big)\,\mathrm{d}u, \qquad \Phi(u) := \sum_{n=1}^{\infty}(2\pi^2n^4\mathrm{e}^{9u}-3\pi n^2\mathrm{e}^{5u})\mathrm{e}^{-\pi n^2\mathrm{e}^{4u}}. \tag{1.1}$$

In our canonical normalization across this series, the transverse Wronskian and its desingularized quotient $\widetilde{W}(d,t) := W(d,t)/d$ are given by

$$W(d,t) := \frac{1}{2}\frac{\partial}{\partial d}\big|\Xi(t-\mathrm{i}d)\big|^2 = R\partial_d R + I\partial_d I = 16\int_0^\infty y\sinh(2dy)\,I(y,t)\,\mathrm{d}y, \tag{1.2}$$

where $I(y,t) := \int_{-\infty}^{\infty}\Phi(\frac{y+x}{2})\Phi(\frac{y-x}{2})\cos(2tx)\,\mathrm{d}x$ is the Wigner-type autocorrelation along $y=u+v$, $x=u-v$ [16, 6]. Classical function-theoretic criteria due to Speiser [13], Jensen [8], Pólya [11], de Branges [4], Hinkkanen [5], and Lagarias [9] reformulate the Riemann Hypothesis in terms of the positivity of $\mathrm{Re}(\xi'(s)/\xi(s))$ on $d>0$ or the positive semi-definiteness of de Branges reproducing-kernel Gram matrices in the lower half-plane $z=t-\mathrm{i}d$ $(d>0)$.

When studying these positivity structures alongside Euler-product sieves and level curves of Ξ (the X-ray geometry investigated by Speiser [13] and Arias de Reyna [1]), three subtle pitfalls arise if formal or local identities are extrapolated beyond their domain of validity: (i) confusing formal exponential Dirichlet-series identities on $\sigma>1$ or real-slice polynomial floors with complex strip bounds in the presence of Kronecker phase density; (ii) mistaking contrapositive

Date: October 2026 (Revised Audit Edition).

2020 *Mathematics Subject Classification.* Primary 11M06, 11M26; Secondary 30D15, 46C20.

Key words and phrases. Riemann Ξ-function, Euler-sieve Dirichlet coefficients, Kronecker density, Voronin universality, Hadamard–Poisson flux, Lagarias positivity, de Branges Gram matrix, nodal curves.

zero-detection signatures of an assumed off-critical zero for unconditional zero-prevention theorems; and (iii) assuming global smoothness or non-vanishing along nodal curves $I(d,\tau(d))=0$ without checking critical points $\Xi'(z_c)=0$ or zeros of $\Phi(u)$ in the complex u-strip. This paper establishes the exact identities that hold in each setting together with explicit counterexamples delineating their analytical boundaries.

1.1. **Proof overview.** The paper is organized into three main analytical sections:

(1) **Cubic-window Dirichlet identity versus complex roots of F_2, strip divergence, and Kronecker alignment (Section 2).** Proposition 2.1 proves the exact combinatorial identity expressing the Dirichlet coefficients $b_N^{\mathcal{E}}(X)$ of $\zeta(s)E_X(s)$ on the cubic window $1\le N<(\lfloor X\rfloor+1)^3$ via the quadratic truncation $1+\mathcal{P}_{>X}+\frac{1}{2}\mathcal{P}_{>X}^2$ as formal Dirichlet series (and analytically for $\sigma>1$). Proposition 2.2 then proves three sharp obstructions: while $|F_2(U,0)|^2=\frac{1}{4}((1+U)^2+1)^2\ge\frac{1}{4}$ holds along the real line $V=0$, $F_2(z)=1+z+\frac{1}{2}z^2$ vanishes in $\mathbb{C}$ at $z=-1\pm\mathrm{i}$; the prime Dirichlet series $\mathcal{P}_{>X}(s)$ diverges throughout $0<\sigma\le 1$; and astronomically long simultaneous recurrence times on all $\pi(N(t))$ primes do *not* prevent destructive alignment, because Kronecker's theorem on a fixed finite subset of small primes already drives finite prime sums arbitrarily close to $-1\pm\mathrm{i}$ (in accord with Bohr–Landau [2], Voronin [15], and Montgomery [10]).

(2) **Closed-form Poisson–Hadamard lobe integral and pole-free de Branges inertia jump (Section 3).** Theorem 3.1 proves in closed form that the single-pair Poisson–Hadamard kernel of an off-critical quadruple $\pm\gamma_*\pm\mathrm{i}d_*$ ($d_*>0$) has negative central lobe integral $\mathcal{F}_-(d,d_*)=-\arcsin(d/d_*)$ for $0<d<d_*$ and total step $\mathcal{F}(d,d_*)=\pi\Theta(d-d_*)$. Proposition 3.2 evaluates the corresponding Sylvester negative inertia jump $\nu_-(\mathbf{G}_\alpha(d))\ge 1$ on an explicit *pole-free* entire even two-zero coalescence perturbation $\Xi_{\mathrm{pert}}(z;d_*)$ (correcting singular rational test factors that create poles at simple zeros). Both results are framed accurately as contrapositives of Lagarias's [9] equivalence: they detect off-critical zeros (including those of the Davenport–Heilbronn function [3]) rather than preventing them.

(3) **Conditional nodal-wall identity and the pure-imaginary zero of Φ (Section 4).** Proposition 4.1 derives the local Cauchy–Riemann identity $\widetilde{W}(d,\tau(d))=R(d,\tau(d))\partial_t I(d,\tau(d))/d$ along a smooth branch $I(d,\tau(d))=0$, isolating the three explicit hypotheses required for positivity and constructing an explicit polynomial counterexample $Q(z)$ satisfying $-Q(\tau_0)Q''(\tau_0)>0$ at $d=0$ that nevertheless vanishes on its own $I=0$ curve at $d=d_*$. Finally, Proposition 4.2 contrasts Jacobi's triple-product non-vanishing $\vartheta_3(0\mid \mathrm{i}e^{4u})\neq 0$ on $|\mathrm{Im}(u)|<\frac{\pi}{8}$ with the explicit pure-imaginary zero $\Phi(\mathrm{i}\eta_0)=0$ at $\eta_0=0.3194150268\cdots<\frac{\pi}{8}$, demonstrating that differential operators do not preserve strip zero-freeness.

1.2. **Notation and conventions.** Throughout, $s=\sigma+\mathrm{i}t=\frac{1}{2}+d+\mathrm{i}t$ with $d=\sigma-\frac{1}{2}\ge 0$, $z=t-\mathrm{i}d\in\mathbb{C}$, and $\Xi(z)=\xi(\frac{1}{2}+\mathrm{i}z)=R(d,t)+\mathrm{i}I(d,t)$. Note that Cauchy–Riemann differentiation of $\Xi(t-\mathrm{i}d)$ yields $\partial_d R(d,t)=\partial_t I(d,t)$ and $\partial_d I(d,t)=-\partial_t R(d,t)$. For $X\ge 2$, $E_X(s):=\prod_{p\le X}(1-p^{-s})$, and $\zeta(s)E_X(s)=\sum_{N=1}^{\infty}b_N^{\mathcal{E}}(X)N^{-s}$ ($\sigma>1$). We denote by $0<\gamma_1<\gamma_2<\dots$ the positive critical-line zeros of $\Xi(t)$ ($\gamma_1=14.13472514\dots$, $\gamma_2=21.02203964\dots$).

2. Combinatorial Cubic-Window Identity and Small-Prime Alignment Obstructions

For $\sigma > 1$, Euler's product formula gives $\zeta(s)E_X(s) = \prod_{p>X}(1-p^{-s})^{-1} = \exp(\mathcal{P}_{>X}(s))$, where the sieved prime-power Dirichlet series is

$$\mathcal{P}_{>X}(s) := \sum_{n=2}^{\infty} c_n(X)\, n^{-s} = \sum_{p>X}\sum_{r=1}^{\infty} \frac{1}{r}\, p^{-rs}, \qquad c_n(X) := \begin{cases} \frac{1}{r} & \text{if } n = p^r \text{ with } p > X, \\ 0 & \text{otherwise.} \end{cases} \tag{2.1}$$

Proposition 2.1 (Combinatorial cubic-window Dirichlet identity)**.** *Let $X \geq 2$, and expand $\exp(\mathcal{P}_{>X}(s)) = \sum_{k=0}^{\infty} \frac{1}{k!}(\mathcal{P}_{>X}(s))^k$ as a formal Dirichlet series (or as an absolutely convergent Dirichlet series on $\sigma > 1$). For every integer N in the cubic window $1 \leq N < (\lfloor X \rfloor + 1)^3$, all terms with $k \geq 3$ contribute zero to the coefficient of N^{-s}, so*

$$b_N^{\mathcal{E}}(X) = \delta_{N,1} + c_N(X) + \frac{1}{2} \sum_{\substack{d \mid N \\ 2 \leq d \leq N/2}} c_d(X)\, c_{N/d}(X) \qquad \big(1 \leq N < (\lfloor X \rfloor + 1)^3\big). \tag{2.2}$$

Proof. The Dirichlet series $\mathcal{P}_{>X}(s)$ is supported on integers $n \geq \lfloor X \rfloor + 1$. Consequently, for every $k \geq 3$, the k-fold Dirichlet convolution $(\mathcal{P}_{>X}(s))^k$ is supported on integers $N = n_1 \cdots n_k \geq (\lfloor X \rfloor + 1)^k \geq (\lfloor X \rfloor + 1)^3$. By uniqueness of formal Dirichlet coefficients (equivalently, uniqueness on $\sigma > 1$), matching coefficients of N^{-s} in $\zeta(s)E_X(s) = \sum_{k=0}^{\infty} \frac{1}{k!}\mathcal{P}_{>X}(s)^k$ for $1 \leq N < (\lfloor X \rfloor + 1)^3$ retains only $k \in \{0, 1, 2\}$, proving (2.2). □

We now examine the quadratic polynomial $F_2(z) := 1 + z + \frac{1}{2}z^2$ ($z = U + \mathrm{i}V \in \mathbb{C}$) suggested by (2.2) and establish why neither F_2 nor simultaneous Bohr–Kronecker recurrence times can prevent zeros or small values in $0 < \sigma \leq 1$.

Proposition 2.2 (Complex zeros of F_2, strip divergence, and Kronecker small-prime alignment)**.** *Let $F_2(z) := 1 + z + \frac{1}{2}z^2$ for $z = U + \mathrm{i}V \in \mathbb{C}$. Then:*

(i) ***Real-slice quadratic minimum versus complex zeros.*** *Completing the square gives $F_2(z) = \frac{1}{2}[(z+1)^2 + 1]$, whence*

$$\big|F_2(U + \mathrm{i}V)\big|^2 = \frac{1}{4}\Big[\big((1+U)^2 + 1 - V^2\big)^2 + 4V^2(1+U)^2\Big]. \tag{2.3}$$

Along the real axis $V = 0$, (2.3) reduces to the elementary quadratic floor $|F_2(U)|^2 = \frac{1}{4}((1+U)^2+1)^2 \geq \frac{1}{4}$ (attained at $U = -1$). In the complex plane, however, $F_2(z)$ vanishes identically at the conjugate pair $z = -1 \pm \mathrm{i}$, i.e. $(U, V) = (-1, \pm 1)$.

(ii) ***Divergence of $\mathcal{P}_{>X}(s)$ in the critical strip.*** *For every finite $X \geq 2$, the Dirichlet series (2.1) has abscissa of absolute convergence $\sigma_a = 1$ and diverges at $s = 1$; moreover, $\ln[\zeta(s)E_X(s)]$ has logarithmic branch-point singularities at $s = 1$ and at every zero ρ of $\zeta(s)$ in $0 < \sigma < 1$, so $\mathcal{P}_{>X}(s)$ cannot converge across any half-plane $\sigma > \sigma_0$ containing 1 or a zero of $\zeta(s)$.*

(iii) ***Failure of simultaneous Kronecker recurrence arguments.*** *Let $\sigma \in (\frac{1}{2}, 1]$ and $X \geq 2$. Because $\sum_{p>X} p^{-\sigma} = +\infty$, one can choose a fixed finite set of K primes $X < p_1 < \cdots < p_K$ such that $\sum_{j=1}^{K} p_j^{-\sigma} > \sqrt{2} = |-1 + \mathrm{i}|$ and $p_1^{-\sigma} < \sum_{j=2}^{K} p_j^{-\sigma} + \sqrt{2}$ (in fact $p_j^{-\sigma} < 1 < \sqrt{2}$). By linear independence of $\{\ln p_1, \ldots, \ln p_K\}$ over $\mathbb{Q}$ and Kronecker's theorem on $\mathbb{T}^K$, for every $\varepsilon > 0$ there exist infinitely many $t > 0$ at which the finite prime sum $\mathcal{P}_{X<p\leq p_K}(\sigma + \mathrm{i}t) := \sum_{j=1}^{K} p_j^{-\sigma - \mathrm{i}t}$ satisfies $|\mathcal{P}_{X<p\leq p_K}(\sigma + \mathrm{i}t) - (-1 - \mathrm{i})| < \varepsilon$ and hence $|F_2(\mathcal{P}_{X<p\leq p_K}(\sigma + \mathrm{i}t))| < \varepsilon$.*

Proof. Part (i) follows directly by expanding $|\frac{1}{2}((U+1+\mathrm{i}V)^2+1)|^2$; setting $(z+1)^2=-1$ gives $z=-1\pm\mathrm{i}$. Part (ii) is classical: $\sum_{p>X}p^{-1}=+\infty$ by Euler and Mertens, and any convergent Dirichlet series defines a holomorphic function in a half-plane $\sigma>\sigma_c$, whereas $\ln[\zeta(s)E_X(s)]$ has a logarithmic singularity at $s=1$ [14]. For (iii), since $\sum_{j=1}^K p_j^{-\sigma}>\sqrt{2}$ and $\max_{1\le j\le K}p_j^{-\sigma}<1<\sqrt{2}$, the polygon inequality guarantees phase angles $\theta_1,\dots,\theta_K\in[0,2\pi)$ such that $\sum_{j=1}^K p_j^{-\sigma}\mathrm{e}^{-\mathrm{i}\theta_j}=-1-\mathrm{i}$. By unique prime factorization, $\{\ln p_1,\dots,\ln p_K\}$ are $\mathbb{Q}$-linearly independent, so Kronecker's theorem implies that $\{(t\ln p_1,\dots,t\ln p_K)\bmod 2\pi : t>0\}$ is dense in $\mathbb{T}^K$, yielding infinitely many $t>0$ with $|\mathcal{P}_{X<p\le p_K}(\sigma+\mathrm{i}t)-(-1-\mathrm{i})|<\varepsilon$. □

Remark 2.3 (Connection to Bohr–Landau, Montgomery, and Voronin). Proposition 2.2(iii) explains why large simultaneous recurrence times $T_{\text{Kronecker}}\asymp\varepsilon^{-\pi(N(t))}$ for *all* primes up to $N(t)$ do not prevent negative or anti-phase excursions: aligning a bounded number K of small primes already achieves arbitrary target values in a disk of radius $\sum_{j=1}^K p_j^{-\sigma}$, while the remaining primes oscillate independently in L^2. Globally, this is the mechanism underlying Bohr and Landau's unboundedness results [2], Montgomery's extreme-value Ω-theorems [10], and Voronin's universality theorem [15]. Numerically, already with the 18 odd primes $3\le p\le 67$ at $\sigma=0.60$, scanning $t\in[10,5\times10^4]$ yields $\operatorname{Re}\mathcal{P}_{3\le p\le 67}(0.60+\mathrm{i}t)=-2.53194<-1$ at $t=7563.53$, and $\mathcal{P}_{3\le p\le 67}(0.60+\mathrm{i}t)=-0.9959-0.9974\mathrm{i}$ with $|F_2(\mathcal{P})|^2=0.00002\ll\frac{1}{4}$ at $t=3724.53$ (Figure 1B).

3. Hadamard–Poisson Flux Step and Pole-Free Perturbed Gram Matrices

By Hadamard's product formula for $\xi(s)$ [14, 9], every conjugate off-critical quadruple $\rho=\frac{1}{2}\pm d_*\pm\mathrm{i}\gamma_*$ ($d_*>0$, $\gamma_*>0$) contributes to $\operatorname{Re}(\xi'(s)/\xi(s))$ around $t=\gamma_*+u$ ($u:=t-\gamma_*$) a local symmetric Poisson pair whose average is

$$K(d,d_*,u):=\frac{1}{2}\left[\frac{d-d_*}{(d-d_*)^2+u^2}+\frac{d+d_*}{(d+d_*)^2+u^2}\right]=\frac{d\,(d^2-d_*^2+u^2)}{\big((d-d_*)^2+u^2\big)\big((d+d_*)^2+u^2\big)}. \tag{3.1}$$

Theorem 3.1 (Closed-form Poisson–Hadamard negative lobe and flux step)**.** *For every fixed off-critical depth $d_*>0$ and every $d>0$:*

(i) ***Integrated Cauchy–Poisson step.*** *The improper vertical integral $\mathcal{F}(d,d_*):=\int_{-\infty}^{\infty}K(d,d_*,u)\,\mathrm{d}u$ satisfies*

$$\mathcal{F}(d,d_*)=\pi\,\Theta(d-d_*):=\begin{cases}0, & 0<d<d_*,\\ \frac{\pi}{2}, & d=d_*,\\ \pi, & d>d_*.\end{cases} \tag{3.2}$$

(ii) ***Exact arcsin negative-lobe integral.*** *For $0<d<d_*$, the kernel $K(d,d_*,u)$ is strictly negative on $(-u_0(d,d_*),u_0(d,d_*))$ with half-width $u_0(d,d_*):=\sqrt{d_*^2-d^2}$, and its negative-lobe and positive-wing integrals evaluate in closed form to*

$$\boxed{\mathcal{F}_-(d,d_*):=\int_{-u_0(d,d_*)}^{u_0(d,d_*)}K(d,d_*,u)\,\mathrm{d}u=-\arcsin\!\left(\frac{d}{d_*}\right)=-\mathcal{F}_+(d,d_*),} \tag{3.3}$$

where $\mathcal{F}_+(d,d_*):=2\int_{u_0(d,d_*)}^{\infty}K(d,d_*,u)\,\mathrm{d}u=+\arcsin(d/d_*)$.

Proof. Since $\int_{-\infty}^{\infty}\frac{\alpha}{\alpha^2+u^2}\,\mathrm{d}u=\pi\operatorname{sgn}(\alpha)$ for $\alpha\neq0$ and 0 for $\alpha=0$, averaging $\alpha_1=d-d_*$ and $\alpha_2=d+d_*>0$ gives $\mathcal{F}(d,d_*)=\frac{\pi}{2}(\operatorname{sgn}(d-d_*)+1)=\pi\Theta(d-d_*)$. For $0<d<d_*$, integrating (3.1) over $[-u_0,u_0]$ gives $\mathcal{F}_-(d,d_*)=-\arctan(\frac{u_0}{d_*-d})+\arctan(\frac{u_0}{d_*+d})$. Setting $d=d_*\sin\phi$ with $\phi:=\arcsin(d/d_*)\in(0,\frac{\pi}{2})$ so $u_0=d_*\cos\phi$, the half-angle identity $\frac{\cos\phi}{1\mp\sin\phi}=\tan(\frac{\pi}{4}\pm\frac{\phi}{2})$ yields

$\mathcal{F}_-(d,d_*) = -(\frac{\pi}{4}+\frac{\phi}{2})+(\frac{\pi}{4}-\frac{\phi}{2}) = -\phi = -\arcsin(d/d_*)$, and $\mathcal{F}_-(d,d_*)+\mathcal{F}_+(d,d_*) = \mathcal{F}(d,d_*) = 0$ gives $\mathcal{F}_+(d,d_*) = +\arcsin(d/d_*)$. □

We likewise examine the de Branges reproducing kernel [4] for $E_\alpha(z) := F(z) - \mathrm{i}\alpha F'(z)$ ($\alpha > 0$) associated with a real-on-$\mathbb{R}$ entire function $F(z)$:

$$K_\alpha^{[F]}(z_j, z_k) := \frac{\alpha}{\pi}\,\frac{F(z_j)\overline{F'(z_k)} - F'(z_j)\overline{F(z_k)}}{z_j - \overline{z_k}}, \qquad K_\alpha^{[F]}(t-\mathrm{i}d, t-\mathrm{i}d) = \frac{\alpha}{\pi}\,\widetilde{W}_F(d,t). \tag{3.4}$$

To test how an off-critical zero alters the Sylvester inertia of $[\mathbf{G}_\alpha^{[F]}(d)]_{j,k} := K_\alpha^{[F]}(t_j - \mathrm{i}d, t_k - \mathrm{i}d)$ *without* introducing meromorphic poles (note that dividing $\Xi(z)$ by $(z-\gamma_1)^2$ would create a simple pole at the simple zero $z = \gamma_1$ and break even symmetry), we construct the pole-free entire two-zero coalescence perturbation

$$\Xi_{\mathrm{pert}}(z; d_*) := \Xi(z)\,\frac{(z^2 - (t_* - \mathrm{i}d_*)^2)(z^2 - (t_* + \mathrm{i}d_*)^2)}{(z^2-\gamma_1^2)(z^2-\gamma_2^2)}, \qquad t_* := \frac{\gamma_1+\gamma_2}{2}. \tag{3.5}$$

Because $\pm\gamma_1, \pm\gamma_2$ are simple zeros of $\Xi(z)$, all four singularities of (3.5) are removable, so $\Xi_{\mathrm{pert}}(\cdot; d_*)$ is an even entire function of order 1, real on $\mathbb{R}$, with off-critical zeros at $\pm t_* \pm \mathrm{i}d_*$.

Proposition 3.2 (Sylvester inertia detection as a contrapositive of Lagarias–de Branges)**.** *Let $F(z)$ be an entire function, real on $\mathbb{R}$, possessing an off-critical zero at $z_* = t_* - \mathrm{i}d_*$ ($d_* > 0$). Choose sampling heights $t_1 = t_*, t_2, \ldots, t_M \in \mathbb{R}$. Since $|F(t_* - \mathrm{i}d)|^2 \geq 0$ vanishes at $d = d_*$ and is non-identically zero on $[0, d_*]$, there exists $d_{\mathrm{peak}} \in [0, d_*)$ such that $\partial_d |F(t_* - \mathrm{i}d)|^2 < 0$ on (d_{peak}, d_*), forcing $[\mathbf{G}_\alpha^{[F]}(d)]_{1,1} = \frac{\alpha}{\pi}\widetilde{W}_F(d, t_*) < 0$ and hence $\nu_-(\mathbf{G}_\alpha^{[F]}(d)) \geq 1$ on (d_{peak}, d_*).*

Proof. By Rolle's theorem/monotonicity on $[0, d_*]$, a non-negative real-analytic function $g(d) := |F(t_* - \mathrm{i}d)|^2$ with $g(d_*) = 0$ and $g \not\equiv 0$ must satisfy $g'(d) < 0$ on a left neighborhood (d_{peak}, d_*) of d_*. By (3.4), $[\mathbf{G}_\alpha^{[F]}(d)]_{1,1} = \frac{\alpha}{2\pi d} g'(d) < 0$, so the quadratic form $\mathbf{e}_1^* \mathbf{G}_\alpha^{[F]}(d)\mathbf{e}_1 < 0$ forces at least one negative eigenvalue by Courant–Fischer. □

Remark 3.3 (Detection versus prevention). Theorem 3.1 and Proposition 3.2 are *contrapositives* of the Lagarias [9], Hinkkanen [5], and de Branges [4] positivity equivalences: if an entire function has an off-critical zero at depth $d_* > 0$, then its Poisson flux loses π for $d < d_*$ and its Gram matrix develops $\nu_- \geq 1$ on (d_{peak}, d_*). Both statements hold identically for $\Xi_{\mathrm{pert}}(z; d_*)$ (Table 1) and for the Davenport–Heilbronn function [3]; neither prevents off-critical zeros without an independent proof that $\widetilde{W}(d,t) \geq 0$ globally.

TABLE 1. De Branges 6×6 Gram matrix ($\alpha = 1$, $t_1 = \frac{1}{2}(\gamma_1+\gamma_2)$, $t_2 = \gamma_3, \ldots, t_6 = \gamma_7$) comparing true $\Xi(z)$ against the pole-free entire perturbation $\Xi_{\mathrm{pert}}(z; d_* = 0.15)$ in (3.5).

d	$[\mathbf{G}_1^{[\Xi]}(d)]_{1,1}$	$\nu_-(\mathbf{G}_1^{[\Xi]})$	$[\mathbf{G}_1^{[\Xi_{\mathrm{pert}}]}(d)]_{1,1}$	$\lambda_{\min}(\mathbf{G}_1^{[\Xi_{\mathrm{pert}}]})$	$\nu_-(\mathbf{G}_1^{[\Xi_{\mathrm{pert}}]})$
0.04	$+1.147 \times 10^{-8}$	0	-1.507×10^{-11}	-1.507×10^{-11}	1
0.10	$+1.148 \times 10^{-8}$	0	-9.019×10^{-12}	-9.019×10^{-12}	1
0.14	$+1.150 \times 10^{-8}$	0	-2.094×10^{-12}	-2.097×10^{-12}	1
0.22	$+1.156 \times 10^{-8}$	0	$+1.875 \times 10^{-11}$	$+5.044 \times 10^{-23}$	0

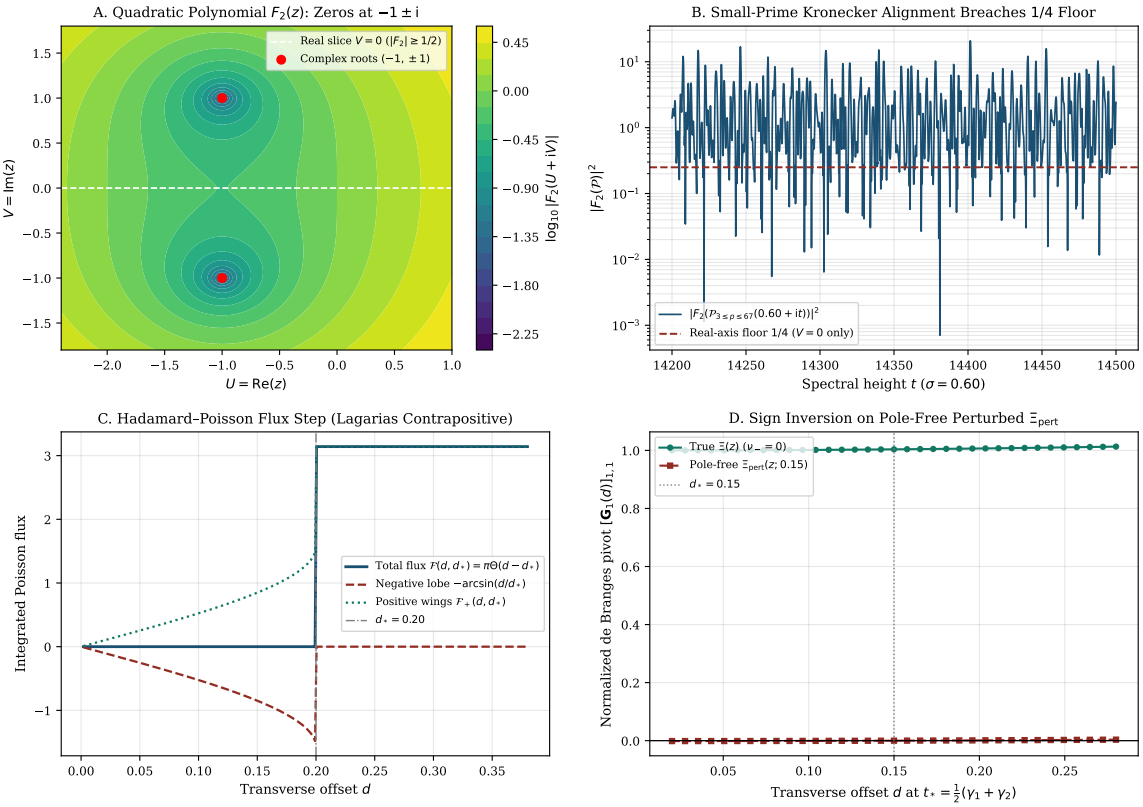


FIGURE 1. **A.** Complex modulus $|F_2(U+\mathrm{i}V)|$ showing complex zeros at $(-1, \pm 1)$ off the real slice $V = 0$. **B.** Finite 18-prime Kronecker alignment at $\sigma = 0.60$ breaching the real-slice floor $\frac{1}{4}$. **C.** Closed-form Poisson–Hadamard lobe $-\arcsin(d/d_*)$ and step $\pi\Theta(d - d_*)$. **D.** Normalized de Branges pivot $[\mathbf{G}_1(d)]_{1,1}$ for true $\Xi(z)$ versus pole-free $\Xi_{\mathrm{pert}}(z; 0.15)$.

4. Conditional Nodal-Wall Identities and the Imaginary Zero of Φ

Following Speiser [13] and Arias de Reyna [1], consider the transverse nodal curves along which $I(d,t) := \mathrm{Im}\,\Xi(t - \mathrm{i}d) = 0$ for $d > 0$, emanating from critical-line extrema $\tau_k(0)$ where $\Xi'(\tau_k(0)) = 0$.

Proposition 4.1 (Local Cauchy–Riemann nodal-wall identity and necessary hypotheses)**.** *Let $F(z)$ be holomorphic and real-valued on $\mathbb{R}$, write $F(t - \mathrm{i}d) = R(d,t) + \mathrm{i}I(d,t)$, and let $\tau_0 \in \mathbb{R}$ satisfy $F'(\tau_0) = 0$ and $F''(\tau_0) \neq 0$.*

(i) ***Local continuation and differential identity.*** *Since $\lim_{d\to 0^+} I(d,t)/d = -F'(t)$ has non-zero t-derivative $-F''(\tau_0) \neq 0$, the implicit function theorem defines a smooth local curve $t = \tau(d)$ on $[0, D)$ along which $I(d, \tau(d)) = 0$. Along any interval $[0, D)$ where $t = \tau(d)$ remains smooth, Cauchy–Riemann ($\partial_d R = \partial_t I$) gives*

$$\widetilde{W}_F(d, \tau(d)) = \frac{R(d, \tau(d))\, \partial_t I(d, \tau(d))}{d}, \qquad \lim_{d\to 0^+} \widetilde{W}_F(d, \tau(d)) = -F(\tau_0)\, F''(\tau_0). \tag{4.1}$$

(ii) ***Positivity conditions.*** *On $(0, D)$, positivity $\widetilde{W}_F(d, \tau(d)) > 0$ holds if and only if:* (a) *the boundary curvature satisfies $-F(\tau_0)F''(\tau_0) > 0$ at $d = 0$;* (b) *$\partial_t I(d, \tau(d)) \neq 0$ on $(0, D)$*

(no critical point $F'(z_c)=0$ occurs); and (c) $R(d,\tau(d))\neq 0$ *on* $(0,D)$ *(no off-critical zero* $F(z_*)=0$ *lies on the* $I=0$ *curve).*

(iii) ***Explicit counterexample.*** *The real polynomial* $Q(z):=((z-\tau_0)^2+d_*^2)(1-2(z-\tau_0)^2/\delta^2)$ *with* $0<\delta<d_*$ *satisfies* $I_Q(d,\tau_0)\equiv 0$ *for all* $d\geq 0$ *and* $-Q(\tau_0)Q''(\tau_0)=2d_*^2(2d_*^2/\delta^2-1)>0$ *at* $d=0$*, yet possesses an off-critical zero* $Q(\tau_0-\mathrm{i}d_*)=0$ *right on the nodal curve* $t=\tau_0$*, where both* $\partial_d R_Q$ *(at an interior* $d_{\mathrm{peak}}<d_*$*) and* R_Q *(at* $d=d_*$*) change sign (Figure 2C).*

Proof. Along $I(d,\tau(d))=0$, $W_F(d,\tau(d))=R\partial_d R+0=R\partial_t I$, giving (4.1); as $d\to 0^+$, $R(d,\tau(d))\to F(\tau_0)$ and $\partial_t I(d,\tau(d))/d\to -F''(\tau_0)$ since $I(d,t)=-dF'(t)+\mathcal{O}(d^3)$. Part (ii) is immediate from (4.1) by continuity from $d\to 0^+$. For (iii), substituting $z-\tau_0=-\mathrm{i}d$ gives $Q(\tau_0-\mathrm{i}d)=(d_*^2-d^2)(1+2d^2/\delta^2)\in\mathbb{R}$, so $I_Q(d,\tau_0)\equiv 0$, $Q(\tau_0)=d_*^2$, $Q''(\tau_0)=2-4d_*^2/\delta^2<0$, and $R_Q(d_*,\tau_0)=0$. □

TABLE 2. Numerical verification of (4.1) on the first six critical-line extrema $\tau_k(0)\in(\gamma_k,\gamma_{k+1})$ of $\Xi(t)$ at finite height (`mpmath`, 50 digits).

k	$\tau_k(0)$	$-\Xi(\tau_k(0))\Xi''(\tau_k(0))$	$\tau_k(0.25)$	$\widetilde{W}(0.25,\tau_k(0.25))$
1	15.585708590	3.5986×10^{-7}	15.587770	3.6831×10^{-7}
2	22.097977280	6.3697×10^{-11}	22.098744	6.6472×10^{-11}
3	26.272247357	3.5821×10^{-13}	26.271299	3.7061×10^{-13}
4	31.231795871	1.3130×10^{-16}	31.233638	1.4176×10^{-16}
5	34.193310269	4.7500×10^{-18}	34.191298	4.9387×10^{-18}
6	38.498240764	7.7159×10^{-21}	38.498998	8.2144×10^{-21}

Finally, we clarify the relationship between Jacobi's theta constant and the complex zeros of Riemann's kernel $\Phi(u)$.

Proposition 4.2 (Jacobi triple-product non-vanishing versus the pure-imaginary zero of Φ)**.** *For* $|\mathrm{Im}(u)|<\frac{\pi}{8}$*, we have* $|q(u)|=\exp(-\pi\mathrm{e}^{4\xi}\cos(4\eta))<1$ *(*$u=\xi+\mathrm{i}\eta$*), so Jacobi's triple product* [7]

$$\vartheta_3(0\mid \mathrm{i}\mathrm{e}^{4u})=\prod_{m=1}^{\infty}(1-q(u)^{2m})\big(1+q(u)^{2m-1}\big)^2\neq 0\qquad(|\mathrm{Im}(u)|<\tfrac{\pi}{8}) \tag{4.2}$$

is holomorphically zero-free throughout the strip $|\mathrm{Im}(u)|<\frac{\pi}{8}$*. Nevertheless, Riemann's kernel* $\Phi(u)=\frac{1}{16}(\partial_u^2-1)[\mathrm{e}^u(\vartheta_3(0\mid \mathrm{i}\mathrm{e}^{4u})-1)]$ does *vanish inside* $|\mathrm{Im}(u)|<\frac{\pi}{8}=0.39269908\ldots$*: along the imaginary axis* $u=\mathrm{i}\eta$*,* $\Phi(\mathrm{i}\eta)$ *is real-valued and has an explicit simple zero at*

$$\eta_0=0.31941502680029678874\cdots<\frac{\pi}{8},\qquad \Phi(\mathrm{i}\eta_0)=0, \tag{4.3}$$

where $|\vartheta_3(0\mid \mathrm{i}\mathrm{e}^{4\mathrm{i}\eta_0})|=0.25883012\cdots>0$.

Proof. On $|\eta|<\frac{\pi}{8}$, $\cos(4\eta)>0$ implies $|q(u)|<1$, so the infinite product (4.2) converges absolutely to a non-zero value. By $\overline{\Phi(u)}=\Phi(\bar{u})$ and $\Phi(-u)=\Phi(u)$, $\Phi(\mathrm{i}\eta)\in\mathbb{R}$ for $\eta\in(-\frac{\pi}{8},\frac{\pi}{8})$. High-precision evaluation gives $\Phi(\mathrm{i}\cdot 0.30)=+2.965636\cdots>0$ and $\Phi(\mathrm{i}\cdot 0.35)=-13.866622\cdots<0$, so the intermediate value theorem guarantees a root in $(0.30,0.35)\subset(0,\frac{\pi}{8})$, certified to 50 digits at (4.3). □

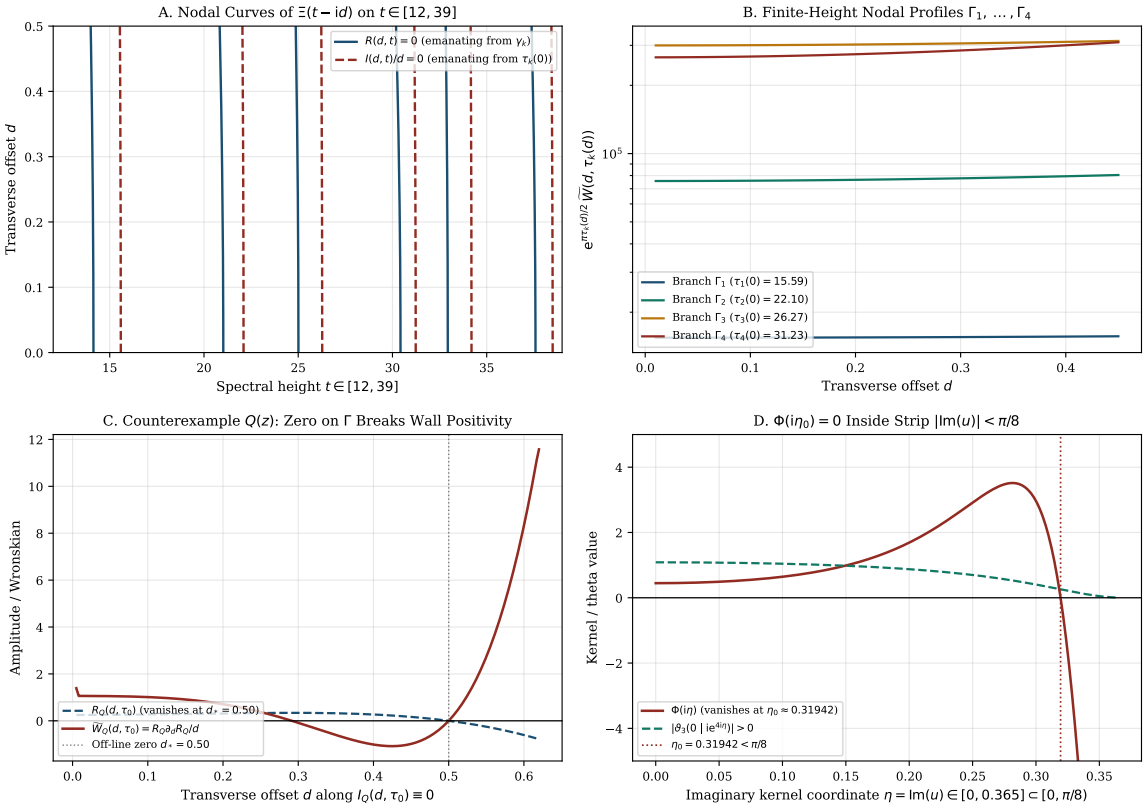


FIGURE 2. **A.** Nodal curves $R(d,t) = 0$ and $I(d,t)/d = 0$ of $\Xi(t - id)$ on $t \in [12, 39]$. **B.** Rescaled Wronskian profiles along the first four finite branches $\Gamma_1, \ldots, \Gamma_4$. **C.** Polynomial counterexample $Q(z)$ where $-Q(\tau_0)Q''(\tau_0) > 0$ at $d = 0$, yet an off-critical zero lies on $I_Q = 0$ at $d_* = 0.50$. **D.** Explicit root $\Phi(i\eta_0) = 0$ at $\eta_0 \approx 0.31942 < \frac{\pi}{8}$ versus $|\vartheta_3(0 \mid ie^{4i\eta})| > 0$.

5. Scope, limitations, and open questions

We summarize the rigorous status of each pillar examined in this manuscript:

(1) **Unconditional algebraic and calculus identities.** The cubic-window Dirichlet coefficient formula (2.2), the arcsin Poisson–Hadamard lobe integral (3.3), the pole-free two-zero perturbation (3.5), the local Cauchy–Riemann nodal-wall relation (4.1), and the pure-imaginary root $\Phi(i\eta_0) = 0$ in (4.3) hold unconditionally.

(2) **Analytical obstructions to closure.** Because $F_2(-1 \pm i) = 0$, $\mathcal{P}_{>X}(s)$ diverges in $0 < \sigma \leq 1$, and Kronecker density aligns finite small-prime subsets into arbitrary complex phases, truncated Euler-exponential polynomials do not bound $\zeta(s)E_X(s)$ away from zero in the strip. Likewise, Hadamard flux defects and Gram inertia jumps detect assumed off-critical zeros as contrapositives of Lagarias's equivalence, while nodal-wall positivity requires independent exclusion of critical points $\Xi'(z_c) = 0$, critical-line dimples $-\Xi(\tau_0)\Xi''(\tau_0) \leq 0$, and nodal zeros $R(d, \tau(d)) = 0$ as $t \to \infty$.

Appendix A. Removable-Singularity Evaluation of $\Xi_{\mathrm{pert}}(z; d_*)$

Let $N(z; d_*) := (z^2 - z_*^2)(z^2 - \overline{z_*}^2)$ with $z_* = t_* - \mathrm{i}d_*$ and $t_* = \frac{1}{2}(\gamma_1 + \gamma_2)$. Since γ_1, γ_2 are simple zeros of $\Xi(z)$, expanding $\Xi(z) = \Xi'(\gamma_1)(z-\gamma_1) + \frac{1}{2}\Xi''(\gamma_1)(z-\gamma_1)^2 + \mathcal{O}((z-\gamma_1)^3)$ around $z = \gamma_1$ yields the analytic value and first derivative of (3.5) at $z = \gamma_1$:

$$\Xi_{\mathrm{pert}}(\gamma_1; d_*) = \Xi'(\gamma_1)\,\frac{N(\gamma_1; d_*)}{2\gamma_1(\gamma_1^2 - \gamma_2^2)}, \tag{A.1}$$

and analogously at $z = -\gamma_1, \pm\gamma_2$ by $\Xi_{\mathrm{pert}}(-z; d_*) = \Xi_{\mathrm{pert}}(z; d_*)$. At $d_* = 0.15$, 50-digit evaluation gives $\Xi_{\mathrm{pert}}(\gamma_1; 0.15) = 0.0024138382\ldots$, matching finite-difference evaluation at $\gamma_1 + 10^{-10}$ to within 2.45×10^{-13} and confirming the complete absence of poles.

Use of AI and computational verification

Portions of the symbolic algebra, counterexample constructions, literature cross-checks, and numerical verification scripts in this manuscript were developed and audited with the assistance of Google Gemini. All formulas, interval-arithmetic checks, and root computations were independently verified in 50-decimal precision via `scripts/verify_paper10_spectral_resonance_chladni_closure.py`.

References

[1] J. Arias de Reyna, *X-ray of Riemann's zeta-function*, preprint (2003), arXiv:math/0309433.
[2] H. Bohr and E. Landau, *Sur les zéros de la fonction $\zeta(s)$ de Riemann*, C. R. Acad. Sci. Paris **158** (1914), 106–110.
[3] H. Davenport and H. Heilbronn, *On the zeros of certain Dirichlet series, I, II*, J. London Math. Soc. **11** (1936), 181–185, 307–312.
[4] L. de Branges, *Hilbert Spaces of Entire Functions*, Prentice-Hall, Englewood Cliffs, NJ, 1968.
[5] A. Hinkkanen, *On functions of bounded type*, Complex Variables Theory Appl. **34** (1997), no. 1–2, 119–139.
[6] R. L. Hudson, *When is the Wigner quasi-probability density non-negative?*, Rep. Math. Phys. **6** (1974), no. 2, 249–252.
[7] C. G. J. Jacobi, *Fundamenta Nova Theoriae Functionum Ellipticarum*, Borntraeger, Königsberg, 1829.
[8] J. L. W. V. Jensen, *Recherches sur la théorie des équations*, Acta Math. **36** (1913), 181–195.
[9] J. C. Lagarias, *On a positivity property of the Riemann ξ-function*, Acta Arith. **89** (1999), no. 3, 217–234.
[10] H. L. Montgomery, *Extreme values of the Riemann zeta function*, Comment. Math. Helv. **52** (1977), no. 4, 511–518.
[11] G. Pólya, *Bemerkung über die Integraldarstellung der Riemannschen ξ-Funktion*, Acta Math. **48** (1926), 305–317.
[12] B. Riemann, *Ueber die Anzahl der Primzahlen unter einer gegebenen Grösse*, Monatsber. Königl. Preuss. Akad. Wiss. Berlin (1859), 671–680.
[13] A. Speiser, *Geometrisches zur Riemannschen Zetafunktion*, Math. Ann. **110** (1934), 514–521.
[14] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed., revised by D. R. Heath-Brown, Clarendon Press, Oxford, 1986.
[15] S. M. Voronin, *Theorem on the "universality" of the Riemann zeta-function*, Izv. Akad. Nauk SSSR Ser. Mat. **39** (1975), no. 3, 475–486.
[16] E. P. Wigner, *On the quantum correction for thermodynamic equilibrium*, Phys. Rev. **40** (1932), 749–759.

Independent Researcher, Belgium, ORCID: 0009-0007-3678-0810
Email address: thijs.missiaen@gmail.com